

Vogle

units 3 & 4

**GEORGIA
POWER**
A SOUTHERN COMPANY

Ninth/Tenth Semi-annual Vogle Construction Monitoring Report

February 2014 • Docket No. 29849



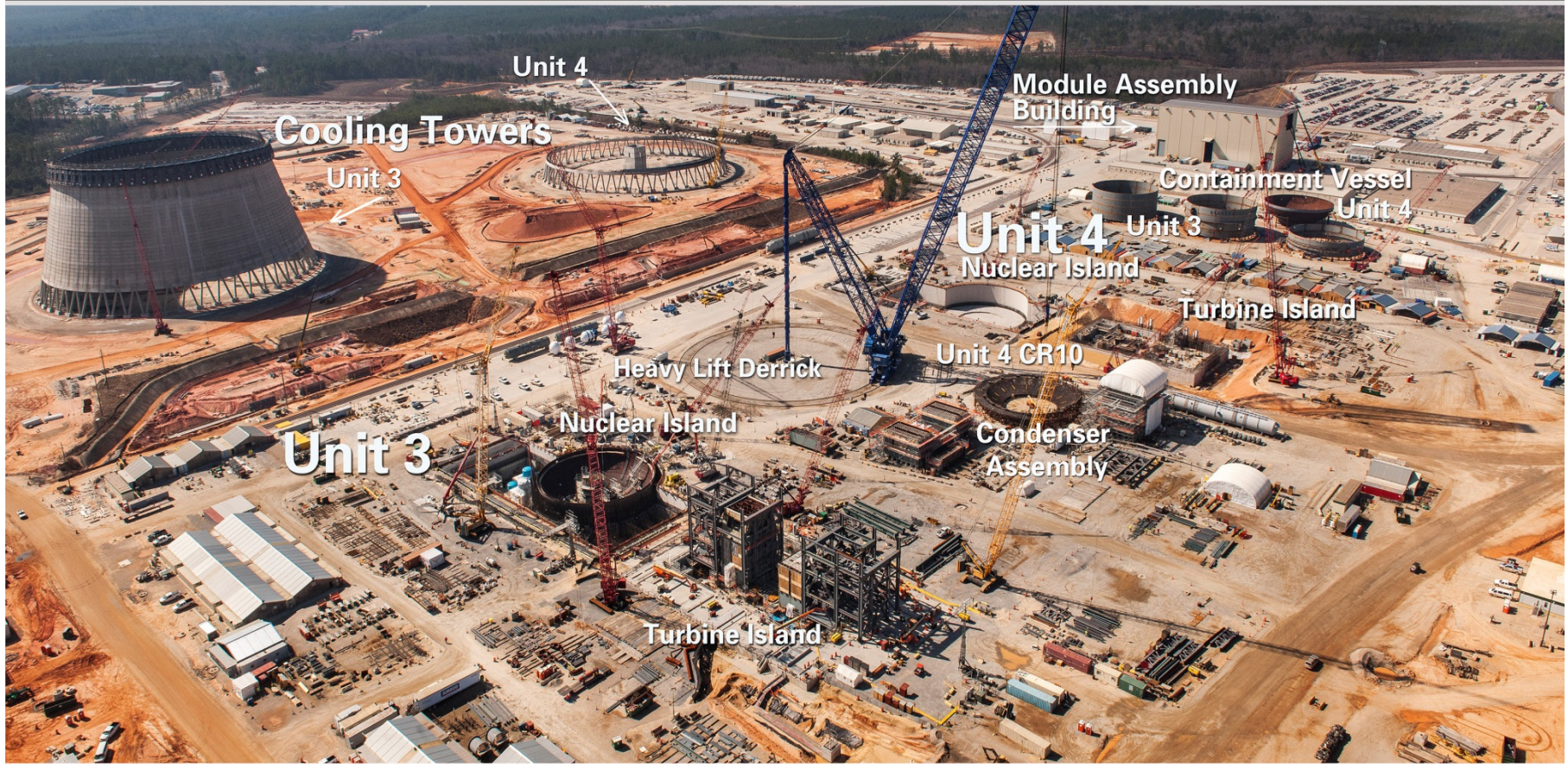
*U.S. Secretary of Energy
Dr. Ernest Moniz and Georgia
Power representatives finalize
the Vogle 3 and 4 loan guar-
antees on Feb. 20, 2014.*

Vogle Units 3 and 4 Ninth/Tenth Semi-Annual Construction Monitoring Report

Table of Contents

	<u>Page</u>
Executive Summary	
I. Introduction	4
II. Highlights	4
Status of the Facility	
III. Safe, Quality, Compliant Construction	9
IV. Transition to Operations	23
V. Facility Capital Cost Overview	26
Responses to Stipulated Questions	30

Vogtle 3&4 - Construction, February 18, 2014



Unit 3 Nuclear and Turbine Islands

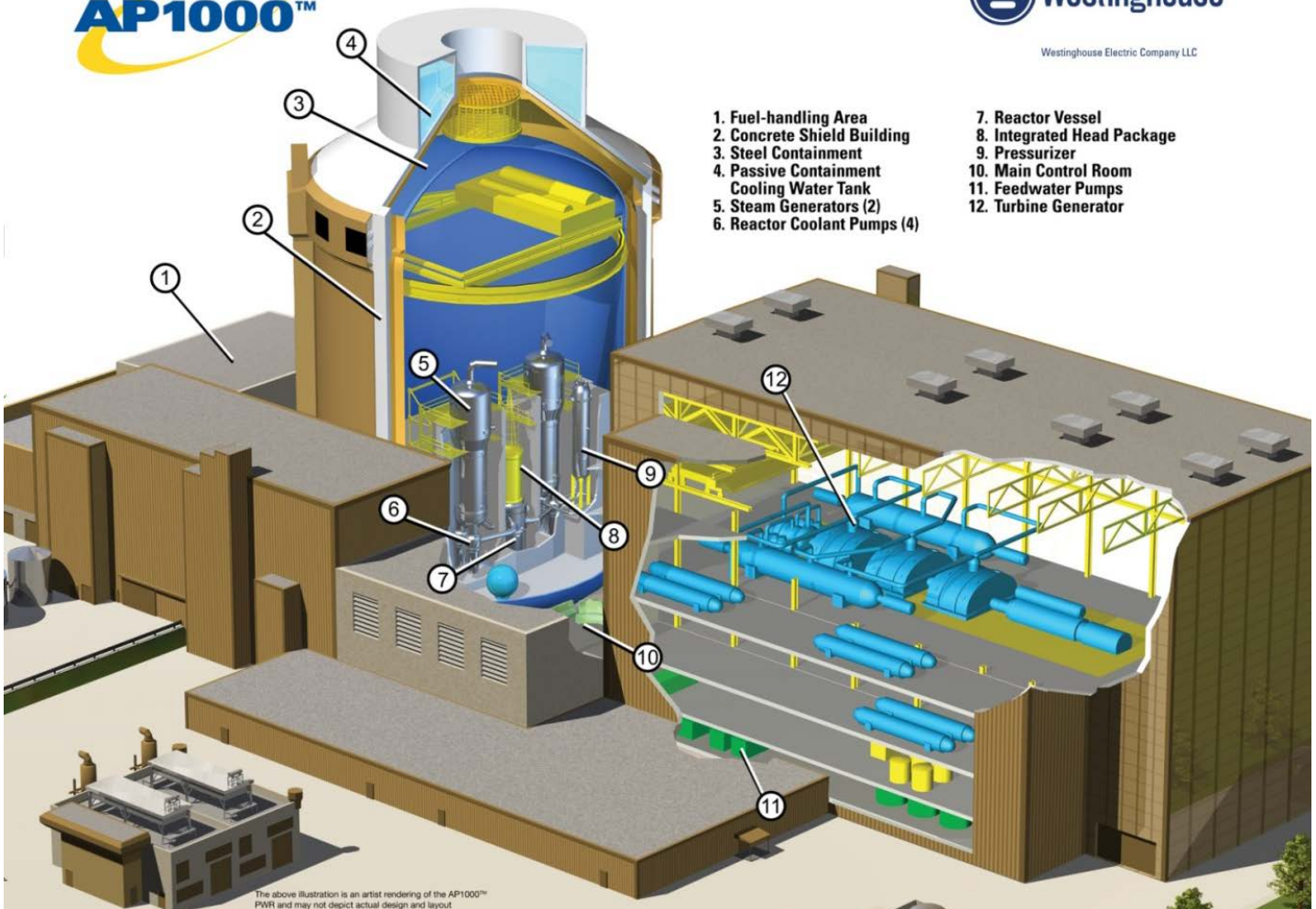


AP1000™



Westinghouse Electric Company LLC

- | | |
|---|----------------------------|
| 1. Fuel-handling Area | 7. Reactor Vessel |
| 2. Concrete Shield Building | 8. Integrated Head Package |
| 3. Steel Containment | 9. Pressurizer |
| 4. Passive Containment Cooling Water Tank | 10. Main Control Room |
| 5. Steam Generators (2) | 11. Feedwater Pumps |
| 6. Reactor Coolant Pumps (4) | 12. Turbine Generator |



EXECUTIVE SUMMARY

I. Introduction

This combined Ninth/Tenth Vogtle Construction Monitoring (“VCM”) Report is submitted by Georgia Power Company (“Georgia Power” or the “Company”) to the Georgia Public Service Commission (“Commission”) for the period of January 1, 2013 through December 31, 2013 (the “Reporting Period”) pursuant to the stipulation adopted by the Commission on September 3, 2013. The Company reports on significant construction progress achieved in 2013 as we advance toward the goal of providing a safe, reliable, clean, and affordable source of electricity to customers that will bring value to Georgia for 60 years or longer. The Vogtle Units 3 and 4 nuclear facility (“Facility”) continues to be well-managed to the current cost and schedule forecasts that have remained unchanged since the Eighth VCM Report filed in February 2013.

In this Ninth/Tenth VCM Report, the Company requests that pursuant to O.C.G.A. 46-3A-7 the Commission verify and approve the expenditures made during the Reporting Period, which total **\$389 million**, as having been made in compliance with the Certificate of Public Convenience and Necessity (“Certificate”). The cumulative Construction and Capital costs for the Facility through this Reporting Period total \$2.6 billion.

II. Highlights

- **The Company continues to demonstrate its uncompromised commitment to safe, quality, and compliant construction of the Facility.** Through its compliance monitoring program, the Company’s oversight efforts are evident in successful Nuclear Regulatory Commission (“NRC”) inspection results and the Company’s preservation of green status in the NRC Construction Reactor Oversight Process.
- **The Total Construction and Capital Cost forecast remains unchanged since the Eighth VCM Report.** The financing costs are reduced by \$91 million. The economic value brought to customers over the life cycle of this Facility continues to increase.
- **The forecasts for Units 3 and 4 commercial operation dates have not changed; they remain fourth quarter 2017 and 2018, respectively.** Fuel load for Unit 3 will begin in mid-year 2017 and the unit will be connected to the grid during start-up testing in third quarter of 2017. Unit 4 activities will occur approximately one year later.
- **Historic construction progress occurred at the Facility during 2013.** Visible progress can be seen in all areas of the Facility including the nuclear islands, onsite module fabrication, turbine islands, cooling towers, and several other permanent Facility buildings.
- **The Company is commencing its transition to operations well ahead of start-up.** Planning is underway and on track for operator training, the initial test program, start-up, and integration into a four unit site as we prepare to connect to the grid and begin commercial operation.

- **The value to customers of completing the Facility remains overwhelmingly positive.**

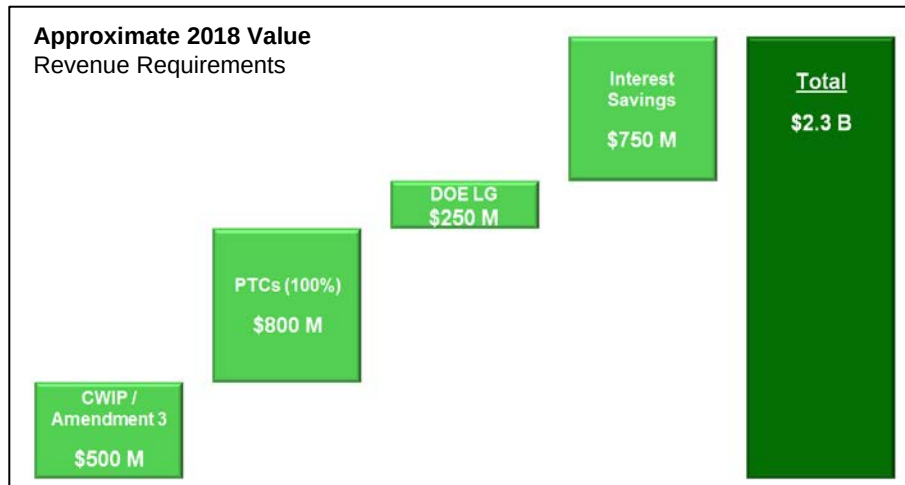
In order to meet future energy needs of our customers, the Commission, whose role is to review and certify necessary capital investments for new plants, together with the Company, recognized the need to bring safe, clean, affordable energy through construction of a new base-load generation source. The economic analysis, performed by the Company and reviewed and approved by the Commission during the Certification proceeding, demonstrated the Facility was the most economic choice when compared to the next best alternative generation source. This analysis recognized fuel savings over the 60 year life-cycle that are associated with investing in nuclear generation. Since that time, the Commission's and the Company's actions have decreased the impact this important capital investment will have on customers, including the Commission's order in the First VCM proceeding to lower the in-service cost of the Facility through inclusion of Construction Work in Progress ("CWIP") in rate base. In a historic moment for the utility industry, on February 20, 2014, the Company and the Department of Energy ("DOE") entered into the DOE Loan Guarantee Agreement, providing an additional \$250 million in financing cost savings for customers over the life of the loan. Lastly, the Company has submitted its applications to the Internal Revenue Service for approximately \$800 million in Production Tax Credits ("PTCs") as a result of reaching the major milestones of placing first nuclear concrete in both Units 3 and 4 in 2013.

Economic analyses by the Company continue to demonstrate that completing this Facility represents the best cost option for our customers by an overwhelming margin. Similar analyses by the Staff of the Commission through the Eighth VCM Report have also demonstrated that completing the Facility represents the best cost option for our customers. Under the Company's current schedule assumptions, completing the Facility provides approximately \$5.0 billion in value today to customers as compared to alternative generation. Even in extended delay scenarios and cost increase sensitivities performed at the Commission's request, the Facility remains the most economic option. The Company notes that the recent short-term natural gas price volatility underscores the need for fuel diversity, especially new nuclear, with its historically stable fuel prices.

The Facility represents at least a 60-year investment in Georgia's critical energy infrastructure. As one of the largest and most capital-intensive infrastructure projects currently underway in the United States, the Facility continues along a path to success that is vital to the local and state economy. Over 5,000 jobs are being created, with approximately 3,200 workers on the Vogtle site today. The Facility has increased the tax base of Burke County as well as the state of Georgia and will continue to do so by creating over 800 full-time, highly skilled and highly paid careers, which will have a compounding effect on the local economy. Georgia's electric supply will continue to have diverse generation, helping to keep prices stable and competitive for many decades to come, thus providing the foundation for a strong, vibrant, and attractive economy.

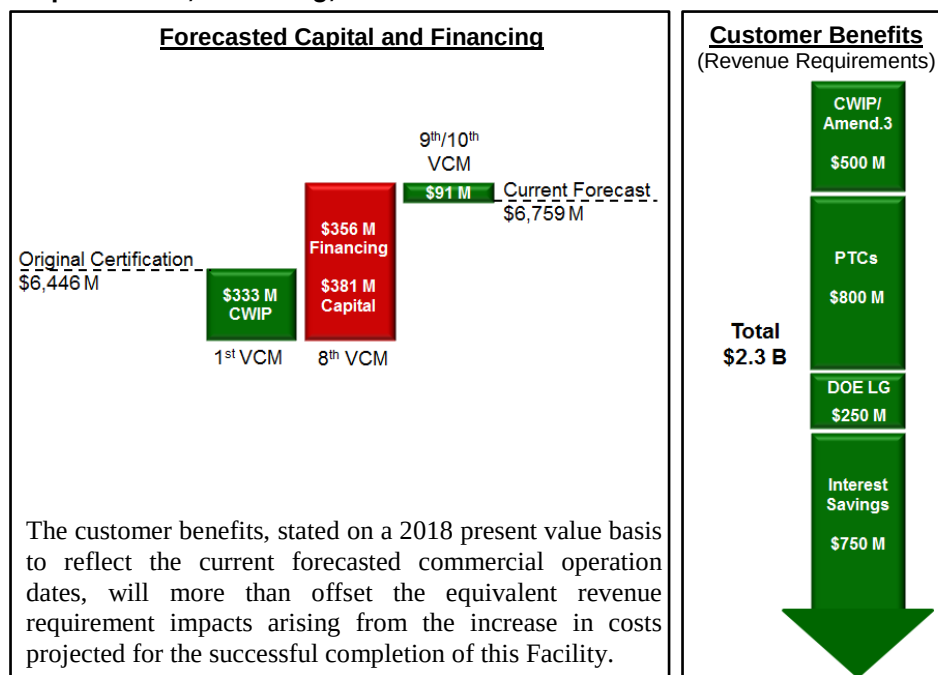
- **The Company is bringing customers over \$2 billion in benefits as a result of constructive regulation, federal government incentives, and a proactive Company financing strategy.** Figure A shows the customer benefits as a result of CWIP in rate base, Amendment No. 3 to the Engineering, Procurement and Construction (“EPC”) Agreement, PTCs, the DOE Loan Guarantee and other interest savings that total \$2.3 billion.

Figure A - Summary of Customer Benefits



- **The Energy Policy Act of 2005 included the DOE Loan Guarantee program and PTCs as “Incentives for Innovative Technologies,” which total \$1.05 billion in customer benefits.** These programs were designed to offset cost impacts that may arise when constructing the first nuclear operating facilities in a generation and will lessen costs to our customers for this **first-of-a-kind construction project**. Figure B, shown in two parts, shows the current forecasted capital cost and financing and separately shows the total \$2.3 billion in customer benefits.

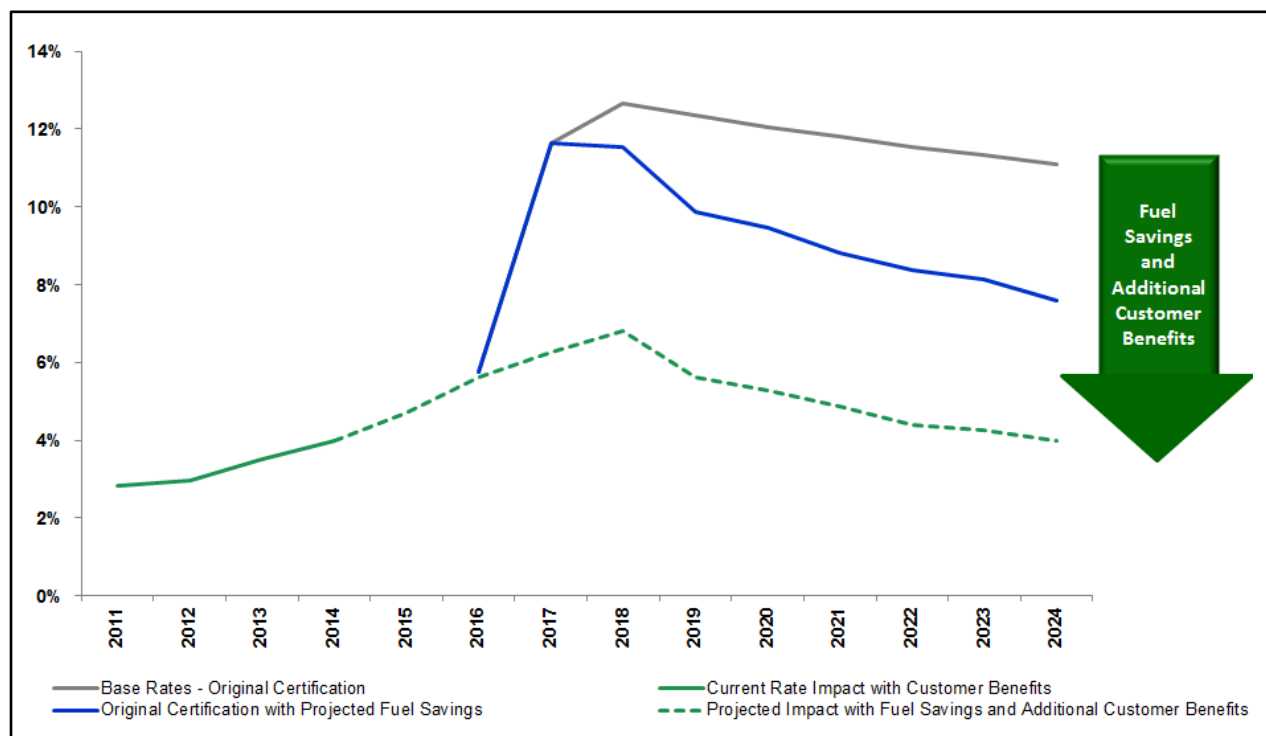
Figure B – Capital Costs, Financing, and Customer Benefits



- Fuel savings associated with investing in nuclear generation, along with the additional customer benefits described above, significantly lessen the overall rate impact to customers at the time the units are put into service.** Without these savings and benefits, the certified capital investment would have resulted in a more significant rate increase at the commercial operation dates (or “CODs”), and rates would have remained substantially higher than those currently projected. One of the key factors in mitigating the initial rate impact to customers at the COD was the Commission’s approval to include CWIP in rate base, which effectively phases in rates over time. The life-cycle fuel savings from nuclear generation significantly lessens the rate impacts in future years as shown in Figure C. In addition, the overall rate impact has been further reduced by other benefits to customers through interest savings, including securing the DOE Loan Guarantee, and the Company’s earned eligibility for PTCs. After considering these life-cycle fuel savings and additional customer benefits, the Facility’s total rate impact is projected to be about 6 to 8 percent.

Figure C shows the benefit to rates resulting from the constructive regulation administered by this Commission together with the federal government incentives for new nuclear construction and the proactive Company financing strategy that have in total provided more value to customers than expected at the time of original Certification.

Figure C – Facility Projected Cumulative Rate Impacts



With 4 percent already in rates, Figure C also shows that **the remaining projected rate impact for customers is only 2 to 4 percent.** Following COD of Units 3 and 4 in the years 2017 and 2018, respectively, the fuel efficiencies from nuclear generation, together with the ongoing customer benefits, are fully expected to put downward pressure on customer rates, as shown above in the downward slope of the graph.

- **The actual expenditures to date are below the original capital cost forecast contemplated at Certification**, representing the Company's oversight and cost control efforts and the value of the EPC Agreement, which has mitigated construction risks of a typical project of this magnitude and complexity. Figure D shows the increase in capital costs since original Certification, which is largely a result of the schedule extension reflected in the Eighth VCM Report from April 2016 and 2017 to the fourth quarter 2017 and 2018 for Units 3 and 4, respectively.

Figure D – Certification versus Current Forecast

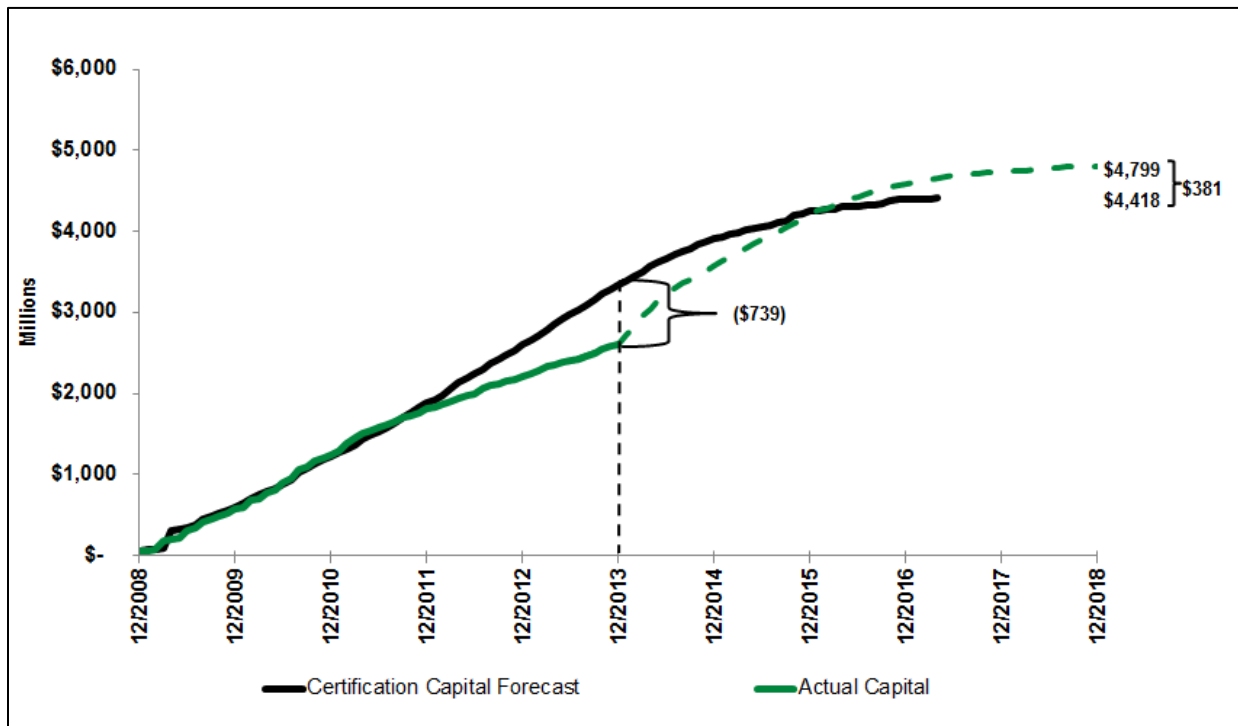


Figure D also demonstrates that the protections included in the EPC Agreement are working to mitigate construction cost risks that were typical in the previous generation of nuclear construction projects. The incremental capital costs as reported in the Eighth VCM Report are primarily associated with federal regulation changes, taxes, operational readiness, owners' quality and compliance, transmission, and legal/environmental permitting. While capital spending has been shifted into later years of construction than that contemplated at the time of original Certification, this does not diminish the value the Facility brings to our customers over at least 60 years.

STATUS OF THE FACILITY

III. Safe, Quality, Compliant Construction

Substantial progress has been made during 2013 at the Vogtle 3 and 4 Facility. Many major milestones were achieved as a result of Westinghouse Electric Company and Stone & Webster (collectively, the “Contractor”) performance with Company oversight in the areas of Construction, Engineering and Procurement. In February 2013, Chicago Bridge & Iron (“CB&I”) acquired Shaw Group and immediately transitioned as Westinghouse Electric Cooperation’s (“Westinghouse”) consortium partner at the Vogtle 3 and 4 site. This change resulted in key leadership changes and the Company has seen overall improvement in the Contractor’s transparency, cooperation and communication as well as execution with a focus on quality and schedule. Together, the Facility management team, which includes Westinghouse, CB&I and the Company, remains committed to meeting the current schedule forecast while maintaining the appropriate nuclear safety focus and culture of compliance associated with all applicable standards.

The Company and Contractor are dedicated to maintaining a culture where the commitment to personnel safety is a core value. With over 3,200 personnel onsite and over 5 million man-hours worked in 2013, the site remains a very safe place to work.

The Company continues to align its oversight efforts to ensure it is most effectively fulfilling its role as the Licensee. During this Reporting Period, the Company successfully expanded and enhanced its compliance monitoring program increasing the number of trained and qualified oversight personnel to perform and document their oversight results as quality surveillances. This process has improved the Company’s ability to verify design and licensing basis requirements are met and use the results to systematically determine the appropriate focus and application of resources. The Company remains in favorable standing with the NRC as indicated by its green status under the NRC’s Construction Reactor Oversight Process. The NRC provides oversight of the Company on a daily basis through its resident inspectors, and in 2013 conducted quarterly inspections, 15 region inspections, and several other site visits. The Company ended 2013 with no open Notices of Violation as reported in the NRC’s fourth quarter inspection report. The NRC also performed several vendor branch inspections at various supplier facilities as a part of their oversight of the Contractor. In March 2013, the NRC reviewed and accepted the Facility’s Integrated Corrective Action Program, which demonstrates the cooperative effort between the Company and the Contractor to effectively implement continuous problem identification and resolution consistent with nuclear industry requirements and standards.

The Company is also submitting Inspections, Tests, Analyses, and Acceptance Criteria (“ITAAC”) Closure Notifications (“ICNs”) to the NRC for verification and ultimate approval prior to fuel load. The Company and the Contractor are working side-by-side in an integrated team setting to ensure alignment through the development of ITAAC Project Plans. This integration fosters a positive working relationship as the Contractor performs the ITAAC activities and the Company submits the final ICNs to the NRC. Eleven ICNs have been successfully submitted to date and the Company expects to submit approximately 34 in 2014.

A. Construction Activities

Nuclear Island

On March 14, 2013, a major construction milestone was completed with the placement of Unit 3 basemat concrete in the nuclear island, which will serve as the foundation for the nuclear island structures. The 41-hour continuous placement consisted of approximately 7,000 cubic yards and required meticulous planning.

In April, the CR10 module was placed in the Unit 3 nuclear island as a support and alignment structure for the containment vessel bottom head (“CVBH”) and the first of several concrete placements was made under CR10 in May. The 909-ton CVBH was subsequently placed in the nuclear island on June 1, 2013. The first concrete placement inside the CVBH was made in early October allowing for placement of the reactor vessel cavity module, CA04, which was placed in early December.

The first pieces of permanent plant equipment, four liquid processing tanks, were placed in the Unit 3 auxiliary building on August 5, 2013. Rebar installation for the Unit 3 auxiliary building walls began in May and the first wall concrete placement was made in mid-August. Currently, all exterior auxiliary building walls have been completed up to elevation 82 ft 6 in. As of December 31, 2013, approximately 1,900 tons of rebar and 8,500 cubic yards of concrete have been installed in the Unit 3 nuclear island to date.

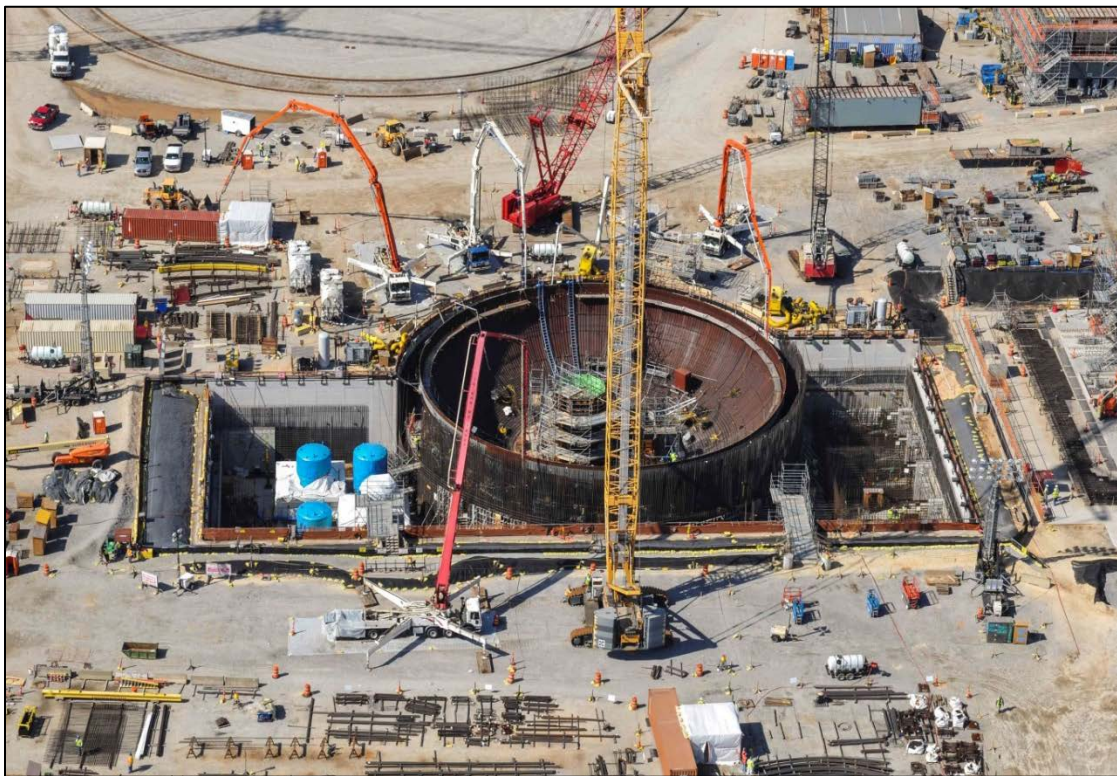
The Unit 4 nuclear island basemat rebar installation began on June 21, 2013. With the successful implementation of lessons learned from Unit 3, basemat rebar installation was successfully completed to support placement of the Unit 4 basemat concrete that began on November 19, 2013. This major milestone was successfully executed in just under 41 hours with no findings from the NRC and was the first major activity to be coordinated through the Contractor Operations Control Center (“OCC”). The Contractor initiated the OCC for this major construction evolution and put it into full operation in January 2014. It is an on-site facility staffed 24 hours a day, 7 days a week to provide a central point for reporting, analysis and resolution of project challenges affecting quality and progress. It is staffed with full-time representatives from all disciplines of the Contractor organization as well as representation from the Company.

Reinforced concrete walls are now being installed above the basemat in the Unit 4 nuclear island. Also utilizing lessons learned from Unit 3, the Unit 4 CR10 module, that forms the cradle for the containment vessel, assembly was completed and lifted into place with the heavy lift derrick in February 2014.

Photo 1 – Unit 3 Nuclear Island in January 2013



Photo 2 – Unit 3 Nuclear Island in February 2014



Turbine Island

Photo 3 – Unit 3 Turbine Island in January 2013



Photo 4 – Unit 3 Turbine Island in February 2014



Vertical construction began in the Unit 3 turbine island with the installation of the first structural steel module, CH80, in late July. Additional structural steel modules, CH81A, CH81C and CH82, have since been installed and will support the turbine deck. The lower and upper shells for the Unit 3 Condenser A have been installed as well as the lower shell for Condenser C. Assembly of the remaining condenser components continues and supports the overall project schedule. The Unit 3 condensate pumps, auxiliary steam supply boiler and flash tank, and other equipment have been installed in the lower elevation of the turbine island. Additional deliveries of the structural steel for the turbine building have arrived to support erection in 2014. Through December 2013, almost 2,300 tons of rebar and 13,500 cubic yards of concrete have been installed in the Unit 3 turbine island.

The foundation slab for Unit 4 turbine island is complete with seven of 10 concrete walls placed for elevation 82 ft 9 in. to 94 ft. Onsite condenser assembly for Unit 4 continues and will meet overall project schedule. As of December 31, 2013, about 1,100 tons of rebar and 7,000 cubic yards of concrete have been installed in the Unit 4 turbine island.

Containment Vessel Assembly

All containment vessel fabrication and assembly is on schedule to meet project needs. Assembly of the Unit 3 containment vessel lower ring and middle ring is essentially complete with attachments and coating work remaining. The assembly of the CVBH for Unit 4 is essentially complete and is expected to be lifted into the nuclear island in second quarter 2014. The Unit 4 lower ring assembly continues to meet overall project schedule.

Photo 5 – Units 3 and 4 Containment Vessel Assembly in February 2014



Cooling Towers

Photo 6 – Unit 3 Cooling Tower in January 2013



Photo 7 – Unit 3 Cooling Tower in January 2014



Veil placement for the cooling tower is now underway and as of December 31, 2013, 26 concrete veil placements were complete that reach 183 ft above ground. More than 5,500 tons of rebar and about 52,000 cubic yards of concrete were placed through December 2013 on the Unit 3 cooling tower.

Installation of the capital columns for the Unit 4 cooling tower is complete and the Contractor is preparing to make the first concrete veil placement.

Balance of Plant

Permanent Facility buildings and structures are progressing well and will meet overall project schedule. Site development is underway to support dewatering of the area for the River Water Intake Structure (Building 326) and concrete structure work is planned to start in the second quarter of 2014. The Pumphouse and Switchgear Building (Building 315) is complete and installation of the mechanical and electrical systems is in progress. The Material Storage Building (Building 322) and the Rotor Storage Building (Building 324) are complete. As of December 31, 2013, the Office Building (Building 301) was about 75 percent complete and the Engineering and Administrative Building (Building 302) was approximately 20 percent complete. Construction has just begun on the Maintenance Support Building as well. In total, there are about 260,000 square ft (6 acres) of above-ground, permanent building construction currently in progress.

Transmission

The construction of new switchyards and modifications to the existing units' switchyards are progressing to meet overall project schedule. All modification work performed in the Unit 1 and 2 switchyard during 2013 was accomplished with no impact to the existing operating units. Design engineering is more than 75 percent complete for the tie lines that will connect to the completed new 230 kV switchyard, which is expected to be commissioned in late 2014. Land preparation and design for a new 500 kV switchyard was also completed and construction is projected to begin in the third quarter of 2014. The land acquisition for the new 55-mile 500-kV transmission line was completed and land clearing began in September to make way for erection of more than 200 tower foundations and structures.

B. Engineering

The Contractor continues to issue Certified for Construction (“CFC”) design documents, to support procurement and construction needs. Timeliness of CFC design document issuance has improved over 2013 but does not consistently meet the objective of document issuance six months prior to the scheduled construction start. As a result of the Company’s continuous analysis and challenge of the Integrated Project Schedule (“IPS”), the Contractor is adding detail to further integrate design engineering activities in the IPS with downstream procurement and construction needs.

During the Reporting Period, the Company provided oversight through the review of Contractor-initiated design changes and completed a comprehensive review of all previously issued nuclear island civil/structural design and design changes to ensure compliance with the license documents. The Company also completed design assessments for significant upcoming construction activities including the CA01 structural module, the shield building, required piping analysis and auxiliary building structural design. These reviews concluded that licensing requirements and applicable codes and standards are being met through the Contractor’s detailed design. Improved Contractor design processes, such as the use of the Structural Review Board to review module and auxiliary building design, have resulted in improvement in the quality of engineering deliverables. Changes in Westinghouse management as well as changes to Contractor engineering processes have resulted in improved interaction with the Company oversight team.

C. Procurement

Fabrication of all major components is either complete or on schedule to meet project requirements. Completion and shipment of several major components during 2013 highlights a significant accomplishment for the Facility. The Company is overseeing fabrication of major Facility equipment all over the world as well as in the United States, and many of the nuclear specific components have been successfully completed. The Company’s subject matter experts in equipment and fabrication processes are observing witness and hold points at vendor facilities, including testing activities, to assure compliance with equipment specifications and design requirements. Technical and quality issues have occurred as nuclear components are fabricated for the first time in three decades for domestic nuclear units; however, to date these issues have been adequately resolved by the Contractor to support the current construction schedule.

Doosan Components

The Unit 3 reactor vessel and head were delivered to the Vogtle 3 and 4 site in April 2013 and the hydrostatic testing has been completed satisfactorily for the Unit 4 reactor vessel. Both Unit 3 steam generators are complete and awaiting preparation for welding to the reactor coolant pump casings later in 2014. Fabrication of the Unit 4 steam generators is progressing with no identified quality issues.

Photo 8 – Unit 4 Steam Generator B at Doosan



The Company resident at the Doosan facility continues to provide oversight for fabrication of these major nuclear components demonstrating the Company's commitment to compliant construction.

Mangiarotti Components

Fabrication completion, successful testing and shipment of the Units 3 and 4 accumulator tanks and the Unit 3 core make-up tanks occurred during this Reporting Period. Documentation package reviews are in progress at the Vogtle 3 and 4 site for these components. The Unit 4 core makeup tanks have been completed but testing results revealed that the volume is slightly lower than requirements. Westinghouse is currently developing a repair plan to bring the volume to the correct specifications. Although delivery may be delayed due to repair work, the Unit 4 core makeup tanks are expected to be shipped with sufficient margin to project need dates.

Photo 9 – Unit 3 Accumulator Tanks and Core Makeup Tanks on the Facility Site



The manufacture of the passive residual heat removal heat exchangers and the pressurizers continues for both units and all remaining Mangiarotti components are expected to ship in 2014 to meet project needs. The Company resident is overseeing the completion of these components.

Other Major Components

All four Unit 3 reactor coolant pumps have been assembled and are currently in the testing phase and the Unit 4 pumps are being assembled. During testing of one of the Unit 3 reactor coolant pumps, an issue was discovered with an internal part. The pump supplier is currently evaluating the test issue and the Company is monitoring resolution. The Company will continue to oversee the testing and fabrication progress to ensure any issues are resolved acceptably.

Also in 2013, Westinghouse identified a material traceability issue with the fabrication of the passive core cooling system squib valves. Affected parts of these valves will be refabricated and replaced. Some redesign was also required and completed as a result of testing results. Retesting is underway and initial results indicate that previously identified design issues have been resolved. The Company is performing quality surveillances to ensure that resolution meets all quality requirements prior to shipment. At this time, there is no anticipated impact from this issue to the overall construction schedule.

Other major equipment deliveries that occurred during 2013 include: Unit 3 reactor coolant surge line piping, Unit 3 and 4 fuel storage racks, Unit 3 and 4 feedwater heaters, Unit 3 main turbine, Unit 3 and 4 moisture separator reheaters, Unit 3 circulating water pump motors, Unit 3 and 4 deaerators, and Unit 3 main step-up transformers.

Photo 10 – Unit 4 Deaerator on the Facility Site



Photo 11 – Unit 3 230 kV Main Step-up Transformers on the Facility Site



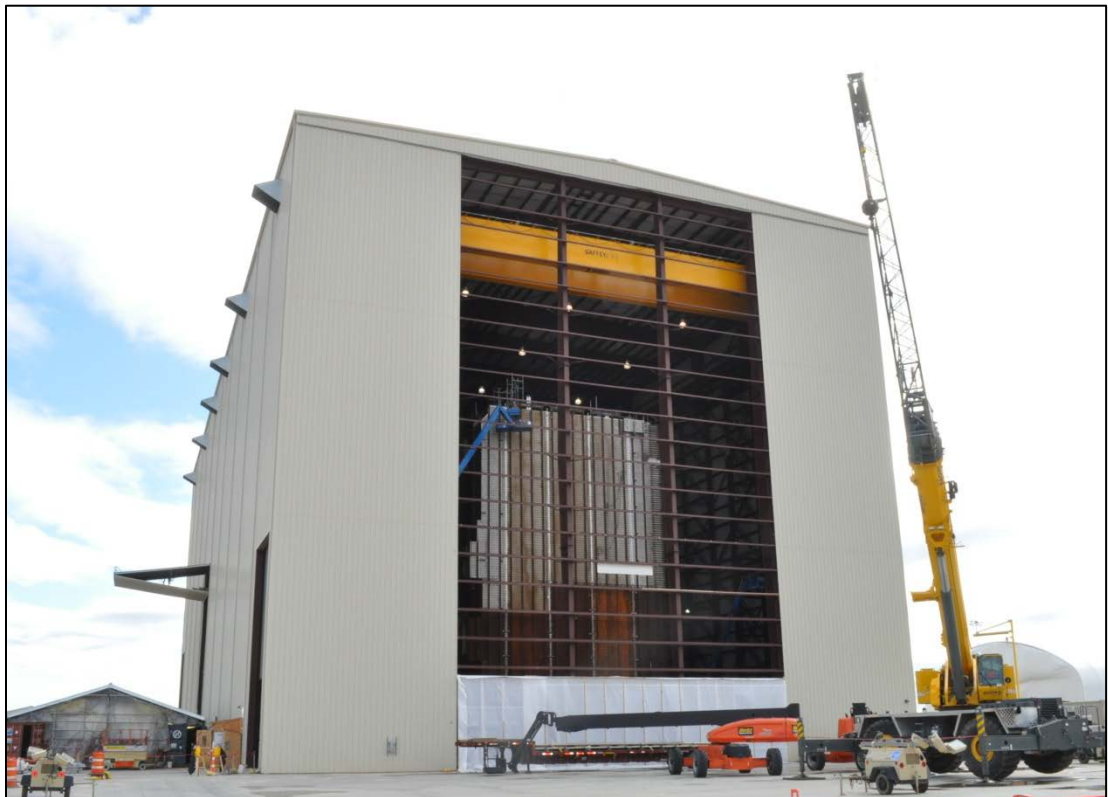
Structural Modules

Quality fabrication of the Unit 3 structural module CA20 has been a primary focus during the Reporting Period. As previously discussed in the Facility Update to the Georgia Public Service Commission filed on September 3, 2013, the Company implemented a stop work order to assure only quality and compliant components are delivered to the Vogtle 3 and 4 site and continues to aggressively monitor the Contractor's review process for release of sub-modules from CB&I-Lake Charles. During the Reporting Period, significant progress has been made in this area, largely as a result of the Contractor decision to deliver all remaining CA20 sub-modules to the site for inspection and repair work. The Contractor established an extension of the Lake Charles facility on the Vogtle 3 and 4 site, Lake Charles-Vogtle, to perform this work that created parallel work paths to mitigate schedule impacts of delayed sub-module deliveries. By the end of October, all CA20 sub-modules had been delivered to the Lake Charles-Vogtle facility. The movement of CA20 inspection and repair work to the Vogtle 3 and 4 site allowed for enhanced supervision and significantly accelerated the release process for assembly in the Module Assembly Building ("MAB").

Photo 12 – Unit 3 CA20 Assembly in the MAB



Photo 13 – MAB Wall Removal to Support Movement of CA20



The Contractor made the decision to fabricate module CA04, the reactor cavity, on the Vogtle 3 and 4 site. CA04 sub-modules were shipped to the Facility site, welded and assembled in the MAB, and the completed CA04 module was placed in the Unit 3 containment vessel on December 4, 2013.

To date, all eight Unit 3 CA05 structural sub-modules have been transported from the CB&I-Lake Charles facility to the Facility site. Eighteen of 47 CA01 sub-modules are complete and 12 have also been shipped from CB&I-Lake Charles and will be inspected in the Lake Charles-Vogtle facility. The Company continues to monitor progress on the fabrication of the remaining Unit 3 major structural sub-modules being fabricated at CB&I-Lake Charles including those that make up CA01 and CA02. The Contractor has decided to expand its supplier base for fabrication of the Unit 3 structural module CA03 and Unit 4 structural modules CA20 and CA01. Many mechanical modules are being fabricated at the CB&I-Lake Charles and fabrication remains on track to meet project needs.

The Newport News Industrial facility is in full production of the shield building structural panels. Production welding began in January 2013 and the first panels were delivered to the Vogtle 3 and 4 site in December 2013.

Photo 14 – Unit 3 Shield Building Panel on Facility Site



IV. Transition to Operations

In parallel to Facility construction, the Company is managing an extensive effort to train personnel and to prepare for component, pre-operational, and start-up testing to ultimately connect to the grid for commercial operation. Associated operational readiness costs above those originally forecast at the time of Certification have been identified and are included in the current capital cost forecast. These costs total \$91 million, representing no change since the Eighth VCM Report.

A. Building the Operational Organization

The Company is strategically building the operational organization with personnel of diverse backgrounds. An accredited training program has been established for licensed plant operator candidates, and as of December 2013, 91 candidates were in different phases of that program. These candidates are taking part in rigorous classroom and simulator training and will be examined by the NRC to obtain their operator licenses.

Photo 15 – AP1000 Simulator in Vogtle 3 and 4 Training Center



Two limited scope simulators have been installed on-site and have proven to be extremely valuable in training operators on the new technology. The Contractor is working to deliver the full Plant Reference Simulator but has experienced challenges in software design and modeling as well as procedure development to support testing and Certification of the simulator by the NRC. The Contractor has developed action plans to address open issues with the simulator design and integration. The first

major milestone towards simulator integration occurred with the start of factory acceptance testing for the Plant Reference Simulator in early February 2014. The Company is providing full time oversight of this testing. In addition, the Company has established a digital systems organization whose sole responsibility is oversight of the quality and compliance of all AP1000 digital systems, to include the Plant Reference Simulator. The Company digital systems organization monitors the Contractor's progress to their action plans and will continue to do so to ensure mitigation of potential impacts to the overall projection schedule through resolution of the current challenges.

Additionally, non-licensed operators, engineering personnel, and maintenance instrumentation and controls technicians have been hired per the staffing plan and are enrolled in comprehensive training programs. The six Maintenance and Technical Training programs recently received initial accreditation from the Institute of Nuclear Power Operations (INPO). Over the past 24 months, the initial accreditation of the 12 required training programs (the six Operations programs in March 2012 and the six Maintenance and Technical programs in November 2013) is the first such accomplishment in the United States in more than 25 years.

The Company is also demonstrating efficient management of resources by using Operational Readiness personnel for augmented construction compliance oversight and startup activities. The use of the Operational Readiness personnel for these activities increases knowledge retention about the Facility's construction in future operations and maintenance activities.

B. Programs, Processes and Procedures

Over 5,000 operations and maintenance procedures are currently being developed by the Company and more than 1,800 procedures have been completed. When possible, existing procedures from the operating fleet are being adopted. Programs are being developed with coordination from fleet experts using a systematic approach that includes detailed schedules and cross-disciplinary management challenge. For those programs with early implementation dates, coordination between the Contractor and the Company will generate detailed program baselines and accurate historical data to be used during operation.

In preparation for initial system turnover during the Initial Test Program ("ITP"), scheduled activities are progressing that will ensure readiness for systems transition (maintenance rule scoping, equipment classifications, maintenance predictive and preventative strategies), design transition (configuration management, digital/software strategy), and licensing program transition (probabilistic risk assessment and risk informed strategies). Additionally, the training necessary to support ITP with trained and qualified personnel is underway, which will support technical program development and generate expertise to support the plant once it is operational.

C. Testing, Turnover, and Startup

The Company and the Contractor have established an integrated ITP organization to jointly plan and implement component testing, pre-operational testing, and startup testing. The ITP organization is working to develop turnover processes from CB&I to Westinghouse and ultimately to the Company. The Contractor will turn the first significant building (Administrative Building) over to the Company in late 2014 that will serve as an early opportunity to exercise the turnover process. Significant work has been accomplished to establish clear roles and responsibilities for the development of procedures to achieve ITP success. Agreements between the Contractor and the Company on division of responsibility have resulted in providing additional intelligence to the overall project schedule.

D. Integrate the Four Unit Site

Collaborative efforts are underway to ensure a smooth transition from a construction site to an operational site with no adverse impacts to the existing Units 1 and 2 while fully supporting the new Units 3 and 4. Preparations for integration into a shared site include, but are not limited to, coordination of physical security changes and emergency planning procedures to remain compliant with all NRC regulations and requirements. Twenty-two sub-strategies have been developed to guide transitional planning in the following five major strategic areas:

- Operational Organization
- Regulatory Margin
- Testing and Turnover Execution
- Multi-Unit Integration
- Sustained Operational Excellence

These strategies are being used to inform and align the organization around the necessary process, program, and procedure work to advance the project through the initial test program and ultimately commercial operation, while achieving all industry standards for excellence in safety and reliability.

V. Facility Capital Cost Overview

A. Status of the Facility Capital Cost

The Total Construction and Capital Cost forecast for the Facility remains \$4.799 billion, which is unchanged from the Eighth VCM Report. The Company is not requesting a change to the certified amount at this time per the stipulation approved by the Commission on September 3, 2013. The cost projected at time of Certification was based on the best available information at the time for items that were variable, such as escalation indices, financing interest rates, and government regulations. Changes to these types of costs over a multi-year construction period, for a complex and first-of-kind project, are not unexpected and were discussed as risks during the Certification proceeding. During the course of normal project implementation, changes will and have occurred. The Facility has experienced cost pressures in the categories as shown below:

Capital Cost Category	Change including escalation (in millions)	Change as a % of Original Certified Capital
Federal Regulation Changes (e.g. cyber and physical security)	\$ 32	0.7
Taxes	50	1.1
Power Block and Support Structure Construction	24	0.5
Operational Readiness, including schedule extension	91	2.1
Owners Quality and Compliance, including schedule extension	153	3.5
Transmission	19	0.4
Legal/Environmental Permit/Misc.	12	0.3
Total Change from Original Certification	\$ 381	8.6

There are certain costs that have arisen from changes in NRC regulations and local government requirements for which the Company and the Contractor have no control. These costs will be spent to meet our obligations to the U.S. government and include items such as changes to NRC requirements for physical and cyber security, the Fitness for Duty program, Fukushima lessons learned, and the Transportation Special Local Option Sales Tax (TSPLOST) implemented by the Central Savannah Region in Georgia.

The protections of the EPC Agreement put into place by the Company and reviewed and found reasonable by this Commission, are effectively working to mitigate and control construction costs. The EPC Agreement provides for some events, such as changes in law and project scope changes, to entitle the Contractor for additional compensation. Cost changes have been acknowledged by the Company as necessary to complete certain activities in a safe, quality and compliant manner such as additional

backfill work, major evolution mock-ups, and implementation of an Integrated Corrective Action Program.

Lastly, there are also additional owners' costs that are necessary to complete the owners' scope of work, and to oversee the Contractor. A large portion of the change in forecasted owners' costs associated with extending the commercial operation dates to the fourth quarter 2017 and 2018 for Units 3 and 4, respectively. The Company's responsibility for ensuring compliance with NRC requirements and the safety of the Facility remains in place through the end of construction and start-up, regardless of the original forecast for commercial operation dates. Other additional costs are necessary for an efficient transition from construction to startup and commercial operations including items such as hardware and software required for plant operations and development of the Probabilistic Risk Assessment.

As the license holder and as co-owner of the Facility, we are committed to fulfilling our role of providing appropriate oversight in every applicable facet of this Facility. The Company recognizes that capital costs that exceed the current certified amount will have to be approved by the Commission before inclusion in rate base.

B. Status of Major Dispute with the Contractor

The Company continues to report on the status of the Contractor's major claim. On August 30, 2013, the District Court for the District of Columbia granted the Owner's motion to dismiss the Contractor claim in that Court, thereby allowing the litigation to proceed in the Southern District of Georgia. The Contractor filed an appeal of the District of Columbia Court's decision on September 27, 2013. While discovery and other pre-trial preparation have commenced and the Company is vigorously asserting its positions and defending against Contractor's claims, the Company expects negotiations with the Contractor to continue with respect to cost and schedule. It is possible that during these negotiations the parties may reach a mutually acceptable compromise of their positions.

As stated in the Eighth VCM Report, the Company has made no provision in the current cost projection for any assumed settlement of the Contractors' claims that are currently subject to litigation in these federal courts. As stated in prior VCM proceedings, if and when any settlement of those claims takes place, the Company will separately present that proposed settlement to this Commission for approval.

C. Customer Benefits

The Company continues to identify and report customer benefits resulting from activities associated with Vogtle Units 3 and 4. In this Ninth/Tenth VCM Report, the value of these projected benefits is reported in 2018 present value to reflect the current forecasted commercial operation dates. These customer benefits can be categorized as follows:

i. CWIP and Amendment No. 3

Approximately \$500 million of customer benefits are associated with the inclusion of CWIP in rate base, as authorized by the Georgia Nuclear Energy Financing Act as well as the Commission. The effectiveness of Amendment No. 3 to the EPC Contract, which shifted more of the EPC costs from market-based indices to fixed escalators, as approved by the Commission, also provided significant cost savings to customers. These customer benefits are included in the Company's capital cost and financing forecast, as discussed in this VCM Report.

ii. Federal Government Incentives

These customer benefits are the product of the Company's effective utilization of federal government nuclear incentives, which are passed along to customers, thereby increasing the overall value of the Facility to customers.

Production Tax Credits (PTCs) - The benefit of the PTCs is earned over an eight year period following the in-service date for each unit and totals approximately \$800 million on a 2018 present value basis. The Energy Policy Act of 2005 provides PTCs for companies that apply for a Combined Construction and Operating Licensing ("COL") by the end of 2008, begin construction on new nuclear reactors prior to 2014 and bring them into service before 2021. The first two requirements have been met for both units with the application for the COLs in 2008, which were subsequently received in 2012, and the placement of first nuclear concrete in 2013. The Company has filed its application for approval with the Internal Revenue Service.

DOE Loan Guarantee - The projected interest cost savings resulting from the DOE Loan Guarantee Agreement are approximately \$250 million on a 2018 present value basis. Pursuant to the loan guarantee program established under Title XVII of the Energy Policy Act of 2005 the Company and the DOE entered into a loan guarantee agreement on February 20, 2014, under which the DOE agreed to guarantee the obligations of the Company under a note purchase agreement ("FFB Note Purchase Agreement") among the DOE, the Company, and the Federal Financing Bank ("FFB") and a related promissory note ("FFB Promissory Note"). The FFB Note Purchase Agreement and FFB Promissory Note provide for a multi-advance term loan facility, under which the Company received an initial \$1 billion draw, and may make additional borrowings through the FFB for an aggregate amount up to the lesser of 70 percent of the total eligible costs, or \$3.46 billion.

This marks the first time the DOE has granted a loan guarantee to encourage new nuclear power construction through its Title XVII Loan Guarantee Program that was designed to help offset potential, up-front cost increases that may occur in a new nuclear facility construction project.

Photo 16 – Commemoration of the DOE Loan Guarantee Agreement



iii. Interest Savings

In addition to the DOE Loan Guarantee, the Company has been able to capture debt cost savings of approximately \$750 million, on a 2018 present value basis, through its proactive financing strategy resulting from lower interest rates contained in the Company's current and projected cost of capital versus the rates assumed during the original Certification proceedings. The majority of these interest savings have already been put into place through debt issuances made through this Ninth/Tenth VCM Reporting Period. Current market rates are below current forecasted rates, so the ultimate savings could be even greater. The interest savings lower financing costs on all Company investments, including the Facility and are being passed on to customers through lower base rate revenue requirements during the Facility's construction period, which includes the NCCR tariff, as well as into the Facility's operating life.

RESPONSES TO STIPULATED QUESTIONS

As agreed in the Stipulation that was incorporated into the Certification Order, the Company responds below to the 15 specified items in the order in which they appear in Section 2(d)(1-15) of the Stipulation. In this Ninth/Tenth VCM Report, and in accordance with the Commission's Order on the Eighth VCM Report, the Company requests that it work with the Public Interest Advocacy Staff to determine items that have reached their completion date or are no longer subject to an appeal process. The request for an item to be deleted in future VCM reports has been noted within each such response herein, as applicable.

1. The reasons for any additional change in the estimated costs of the units since the process began.

There has been no change to the Total Construction and Capital Cost of the Facility since the Eighth VCM Report. However, Total Construction Schedule Financing costs have decreased since the Eighth VCM Report, due to lower cost of capital rates and changes in the timing of project cash expenditures. The current cost and forecast reports are provided in Tables 1.1 and 1.1a.

Changes to the estimated costs of the Facility since original Certification are discussed in the Executive Summary and the Status of the Facility sections of this Ninth/Tenth VCM Report.

Table 1.1

Vogtle 3&4 Facility
Georgia Power Company Cost
Project To Date
Through Period Ending December 31, 2013

	Total Capital Cost and Financing				Facility To Date Cost			
	Certified Cost (\$ millions)	Total Current Forecast (\$ millions)	Variance (\$ millions)	Footnote	Budget To Date (4) (\$ millions)	Actual To Date (\$ millions)	Variance (\$ millions)	Footnote
Construction & Capital Cost								
EPC Base								
Fixed Semi Annual Escalation	1,978	1,976	-2		1,264	1,244	-21	
Indexed Escalation	468	470	2		126	114	-12	
Other Fixed Escalation	670	674	4		607	597	-10	
Total EPC Base	3,116	3,121	4		1,998	1,955	-43	
EPC Escalation								
Fixed Semi Annual Escalation	431	355	-76		140	137	-4	
Indexed Escalation	142	117	-25		15	13	-2	
Other Fixed Escalation	108	110	2		84	81	-3	
Total EPC Escalation	681	582	-99		240	230	-9	
Quality Assurance, Compliance and Operations & EPC Scope Change	507	930	423	1	369	369	0	
Ad Valorem	111	159	48		17	17	0	
Test Fuel Offsets	-34	-49	-15		0	0	0	
Transmission Interconnection	37	56	20		30	29	-1	
	621	1,096	476		416	415	-1	
Total Construction & Capital Cost	4,418	4,799	381		2,653	2,599	-54	
Construction Schedule Financing								
Return on CWIP in Rate Base	1,545	1,851	306	2	441	438	-3	
AFUDC - Accrued through Dec 2010	111	91	-20		91	91	0	
Return on Unamortized AFUDC Balance	39	18	-21		15	15	0	
Total Construction Schedule Financing	1,695	1,960	265	3	548	545	-3	
Total Capital Cost and Financing	6,113	6,759	646		3,201	3,144	-57	
Other Capital Cost								
Certification & Independent Evaluator Fees	0	0	0		0	2	2	
Construction Monitor	0	5	5		2	2	0	
Total Other Capital Cost	0	5	5		2	4	2	

Footnotes

1. See Item 9 for additional information regarding the Total Current Forecast for Quality Assurance, Compliance and Operations & EPC Scope Change Category since the Eighth VCM Report.
2. The Total Current Forecast for Return on CWIP in Rate Base includes \$25 million of AFUDC accrued on CWIP above the certified cost.
3. The Total Current Forecast for Total Construction Schedule Financing decreased due to timing of EPC milestones and the lower projected cost of capital.
4. The Budget to Date includes actual costs through the previously filed report, plus budgeted costs through the Ninth/Tenth VCM Reporting Period.

Table 1.1.a (Trend)

Vogtle 3&4 Project
Georgia Power Company Cost Forecast
Through Period Ending December 31, 2013

	Certified Cost (\$ millions)	Jun 2009 Forecast (\$ millions)	Dec 2009 Forecast (\$ millions)	Jun 2010 Forecast (\$ millions)	Dec 2010 Forecast (\$ millions)	Jun 2011 Forecast (\$ millions)	Dec 2011 Forecast (\$ millions)	Jun 2012 Forecast (\$ millions)	Dec 2012 Forecast (\$ millions)	Dec 2013 Forecast (\$ millions)
Construction & Capital Cost										
EPC Base										
Fixed Semi Annual Escalation	1,978	1,978	1,976	1,976	1,976	1,976	1,976	1,976	1,976	1,976
Indexed Escalation	468	468	470	470	470	470	470	470	470	470
Other Fixed Escalation	670	670	674	674	674	674	674	674	674	674
Total EPC Base	3,116	3,116	3,121	3,121	3,121	3,121	3,121	3,121	3,121	3,121
EPC Escalation										
Fixed Semi Annual Escalation	431	431	336	336	337	344	343	353	355	355
Indexed Escalation	142	142	142	142	142	119	118	120	117	117
Other Fixed Escalation	108	108	109	109	109	110	110	111	110	110
Total EPC Escalation	681	681	586	587	589	573	572	585	582	582
Quality Assurance, Compliance and Operations & EPC Scope Change	507	507	576	589	582	675	675	727	930	930
Ad Valorem	111	111	111	111	111	111	111	125	159	159
Test Fuel Offsets	-34	-34	-34	-34	-34	-60	-60	-60	-49	-49
Transmission Interconnection	37	37	37	40	40	40	40	41	56	56
	621	621	689	706	699	766	766	833	1,096	1,096
Total Construction & Capital Cost	4,418	4,418	4,395	4,414	4,408	4,460	4,459	4,539	4,799	4,799
Construction Schedule Financing										
Return on CWIP in Rate Base	1,545	1,507	1,505	1,546	1,553	1,524	1,516	1,552	1,942	1,851 *
AFUDC - Accrued through Dec 2010	111	97	99	99	91	91	91	91	91	91
Return on Unamortized AFUDC Balance	39	32	33	33	31	19	19	18	18	18
Total Construction Schedule Financing	1,695	1,636	1,637	1,678	1,675	1,635	1,626	1,662	2,051	1,960
Total Capital Cost and Financing	6,113	6,054	6,032	6,092	6,083	6,095	6,085	6,201	6,850	6,759
Other Capital Cost										
Certification & Independent Evaluator Fees	0	0	0	0	0	0	0	0	0	0
Construction Monitor	0	5	5	5	4	4	4	4	4	5
Total Other Capital Cost	0	5	5	5	4	4	4	4	4	5

Note: No reforecast was filed for June 2013.

Details may not add to totals due to rounding.

*The total current forecast for Return on CWIP in Rate Base includes \$25 million of AFUDC accrued on CWIP above the certified cost.

2. A description of any cooperative actions between other builders of nuclear units in the southeast to address labor, crafts, engineering and management requirements.

There has been no change in the status of this item since the last reporting period.

The Company recommends removing this item in future VCM reports.

3. An explanation of how the indices used in the EPC contract are tracking.

There has been no change in the status of this item since the last reporting period.

The Company recommends removing this item in future VCM reports.

4. Any updated estimate of onsite spent fuel storage costs, including the costs for dry storage of spent fuel for an extended period of time after shutdown, and any updated calculation of spent fuel storage costs assuming Yucca Mountain is never available.

There has been no change in the status of this item since the last reporting period.

The Company recommends removing this item in future VCM reports.

5. The status of the Company's loan guarantee application at the Department of Energy and to the extent that application is granted, then the Company shall also report on the impact it has or would have on the final expected in-service cost of the units.

Pursuant to the loan guarantee program established under Title XVII of the Energy Policy Act of 2005 the Company and the DOE entered into a loan guarantee agreement on February 20, 2014, under which the DOE agreed to guarantee the obligations of the Company under a note purchase agreement ("FFB Note Purchase Agreement") among the DOE, the Company, and the Federal Financing Bank ("FFB") and a related promissory note ("FFB Promissory Note"). The FFB Note Purchase Agreement and FFB Promissory Note provide for a multi-advance term loan facility, under which the Company may make term loan borrowings through the FFB.

This marks the first time the DOE has granted a loan guarantee to encourage new nuclear power construction through its Title XVII Loan Guarantee Program that was designed to help offset potential, up-front cost increases that may occur in a new nuclear facility construction project. The projected interest cost savings resulting from the Loan Guarantee Agreement are approximately \$250 million on a 2018 present value basis. These savings will be realized to customers through a decrease in base rate revenue requirements including the NCCR tariff. See the Customer Benefits section of this report for additional information on the DOE Loan Guarantee savings to customers.

The current estimated impact of the DOE loan guarantee on this Facility is reflected in financing costs in Tables 1.1, 1.1a, and 8.1. The portion of the estimated net benefits of the loan guarantee that is allocated to the Facility has been reflected in the Total Current Forecast. Since the DOE Loan Guarantee Agreement is now in place, the impact of the DOE loan guarantee is also now reflected in the economic analysis in Item 14.

The Company recommends removing this item in future VCM reports.

6. Whether the Company is using trust preferred financing and the impact it has or would have on the expected in-service cost of the units.

There has been no change in the status of this item since the last reporting period.

The Company recommends removing this item in future VCM reports.

7. The extent to which the Company is using short term debt and the impact it has or would have on the expected in-service cost of the units.

There has been no change in the status of this item since the last reporting period.

The Company recommends removing this item in future VCM reports.

8. An update of the estimated in-service cost and projected date of commercial operation of both units.

The estimated in-service cost of both units is shown in Table 8.1. As discussed in earlier sections of this report, there has been no change to the Total Construction and Capital Cost of the Facility since the Eighth VCM Report. Total Construction Schedule Financing costs have however decreased due to lower cost of capital rates and changes in the timing of cash expenditures.

The Company's forecast for commercial operation dates remain fourth quarter 2017 and fourth quarter 2018 for Units 3 and 4, respectively.

Table 8.1

**Vogle 3&4 Facility
Capital Expenditure and Financing Cost Report
Through Period Ending December 31, 2013**

TOTAL CAPITAL COST AND FINANCING

	Construction & Capital Balance (\$ millions)	Financing Balance (\$ millions)	Total Project Balance (\$ millions)	Certified Projected Project Balance (\$ millions)	Variance Dollars (\$ millions)	Variance Percent
Jun 30, 2009	314	9	323	349	-26	-7%
Dec 31, 2009	583	25	608	635	-27	-4%
Jun 30, 2010	994	54	1,048	949	99	10%
Dec 31, 2010	1,246	91	1,337	1,341	-4	0%
Jun 30, 2011	1,580	144	1,724	1,716	8	0%
Dec 31, 2011	1,808	208	2,016	2,137	-121	-6%
Jun 30, 2012	2,001	278	2,279	2,590	-311	-12%
Dec 31, 2012	2,210	361	2,572	3,057	-485	-16%
Jun 30, 2013	2,406	446	2,852	3,569	-717	-20%
Dec 31, 2013	2,599	545	3,144	4,063	-919	-23%
Jun 30, 2014	2,831	644	3,475	4,557	-1,082	-24%
Dec 31, 2014	3,196	762	3,958	4,963	-1,005	-20%
Jun 30, 2015	3,529	889	4,418	5,304	-886	-17%
Dec 31, 2015	3,950	1,039	4,989	5,676	-687	-12%
Jun 30, 2016	4,233	1,196	5,429	5,915	-486	-8%
Dec 31, 2016	4,459	1,382	5,841	6,049	-208	-3%
Jun 30, 2017	4,629	1,560	6,189	6,113	76	1%
Dec 31, 2017	4,728	1,771	6,499	6,113	386	6%
Jun 30, 2018	4,777	1,857	6,634	6,113	521	9%
Dec 31, 2018	4,799	1,960	6,759	6,113	646	11%

9. A description of all major sources of changes (both increases and decreases) to the in-service cost and sources of change in commercial operation dates, if any.

There is no change to the forecasts for the in-service cost nor the commercial operation dates since the Eighth VCM Report.

10. The status of the Company's COL applications at the Nuclear Regulatory Commission.

As reported in the Sixth VCM Report, the NRC approved the Unit 3 and Unit 4 COLs on February 9, 2012 and subsequently issued the COLs on February 10, 2012.

The Company recommends removing this item in future VCM reports.

11. The status of all other significant permits and licenses required from other governmental agencies.

All other required permits and licenses have been approved or are on track to be approved to meet construction need dates as shown in the Permits Update filed monthly with the Commission. There has been no change in the status of this item since the January 2014 Monthly Status Report was filed.

The Company recommends removing this item in future VCM reports.

12. The status of procurement, engineering, fabrication, transportation and erection of major equipment.

The status of procurement, engineering, fabrication, transportation and erection of major equipment is reported in the Status of the Facility section of this report.

13. The status of transportation links for heavy forgings and modules.

Forging activities have been completed and all required forgings have been transported for major components to begin fabrication. Fabrication of equipment from these forgings is progressing on schedule and deliveries of all major components are scheduled to meet construction need dates. No additional heavy forgings were transported to manufacturing facilities during the reporting period. Structural module deliveries are discussed in the Status of the Facility Section of this report.

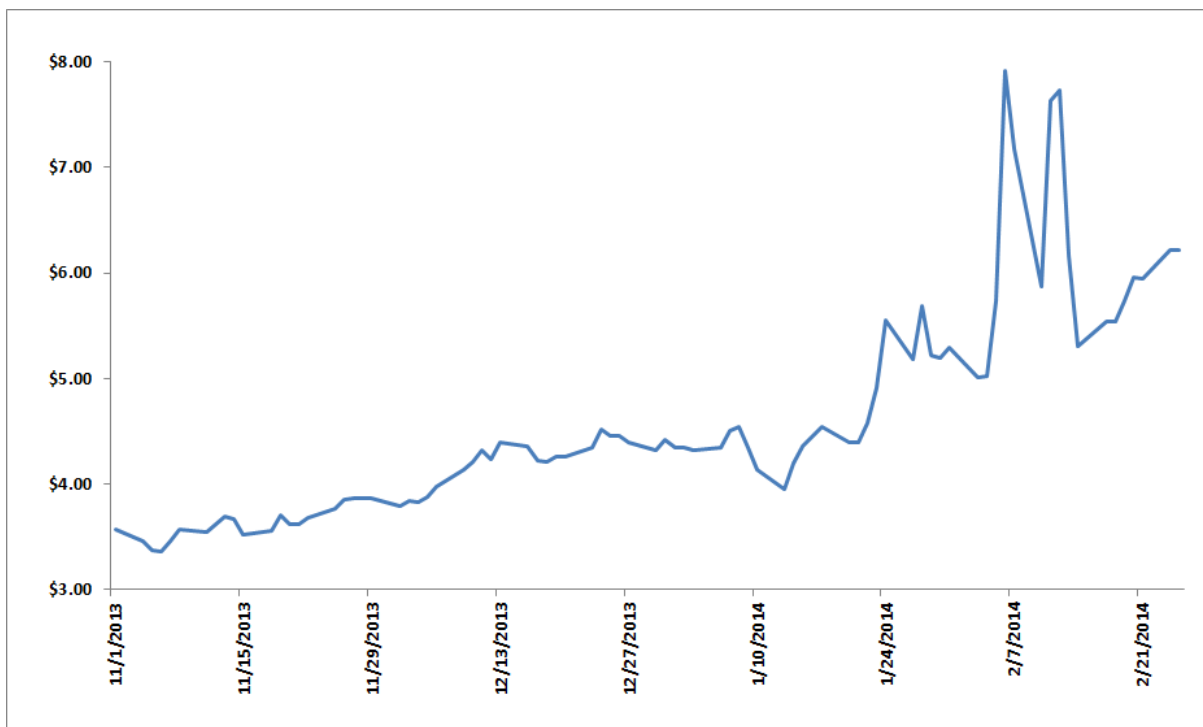
The Company recommends removing this item in future VCM reports with the understanding that module delivery status will be provided in Item 12.

14. An updated comparison of the economics of the certified project to other capacity options.

The relative economic value of the Facility can be determined by comparing the costs associated with completing, operating, and maintaining the Facility over its expected 60-year useful life with the costs to build, operate, and maintain a combined cycle (“CC”) natural gas alternative, which is the next most viable generation alternative, over a comparable time period. The economic analysis performed for this Ninth/Tenth VCM Report has relied on the methodologies used in all previous economic evaluations conducted in Docket Nos. 27800 and 29849.

The economic evaluation presented in this Ninth/Tenth VCM Report includes updates of all major underlying planning assumptions including fuel forecasts, load forecasts, and new generation technology costs. Consistent with the original Certification filing and all previous VCM reports, a range of planning scenarios was used to evaluate the possible impacts of varying fuel prices and carbon costs. The Company identified four distinct, useful views of future North American natural gas supply and demand conditions in its 2014 fuel forecast – “Low”, “Moderate”, “Restrained” and “High”. The Company notes that despite lower long term natural gas price forecasts, largely due to shale gas developments, the gas markets are still experiencing significant short term price volatility as recently as the last few months due to the extreme cold weather. Figure E demonstrates this recent, short term volatility.

Figure E – Henry Hub Natural Gas Prices – Winter Months



In addition to price volatility in the supply regions, there has also been extreme volatility in delivered prices to natural gas markets across the eastern half of the United States this winter. This reflects transportation constraints to areas where gas is needed. For example, on January 22, 2014, daily gas prices reached over \$100/MMBtu in some of the eastern regions. While the effects of short-term volatility are not directly reflected in our long-term forecasts or the Vogtle economics, they are felt by our customers. This underscores the need for fuel diversity, especially new nuclear, with its historically stable fuel prices.

The carbon cost scenarios remain similar to those in the Eighth VCM and are: “Existing” (\$0), “Moderate” (\$10, beginning in 2018 and escalated), “Substantial” (\$20, beginning in 2022 and escalated).

The estimate of the capital cost to complete the Facility has been updated from the Eighth VCM Report along with pre-COD Operations and Maintenance (“O&M”), post-COD O&M, projected post-COD ongoing capital additions, nuclear fuel, and spent fuel storage cost estimates. Decommissioning costs and the assumed operating characteristics of the Facility have not changed. The long-term marginal financing rates for debt and preferred stock were reviewed but have not been changed from the Eighth VCM Report. The cost of equity has changed to reflect the most recent allowed return as approved in the Company’s 2013 base rate case. It should be noted that these marginal financing costs are higher than the current estimate of embedded average financing costs, which are used in all other references to financing costs in this report. Consistent with the Eighth VCM Report, the current economic evaluation assumes 50 percent of potentially available PTCs. Finally, this economic analysis now includes the expected interest savings of the DOE loan guarantee, which was closed in February 2014.

“Sunk costs” (non-refundable capital and financing costs already incurred or projected to have been incurred as of February 28, 2014) are excluded from this forward-looking analysis. The current forecast of construction and capital costs as shown in Table 1.1, net of sunk costs, is used as the basis to determine “cost to complete.”

The relative economics of the Facility, when compared to the gas-fired CC alternative, vary depending on the assumptions for future fuel prices as well as with the projected carbon costs associated with potential future carbon regulation. Table 14.1 below shows the difference between the lifetime costs of building, operating, and maintaining the gas-fired CC alternative and the Facility, with positive savings meaning the Facility is less expensive to customers than the gas-fired CC alternative. All twelve scenarios show positive benefits to customers for completing and operating the Facility.

Table 14.1

Relative Savings of the Facility versus CC as of February 28, 2014
“Incremental Cost to Complete”

(In 2016 Dollars)

(Net present value of lifetime costs of CC minus the Facility)

Fuel \ CO ₂	Existing CO ₂	Moderate CO ₂	Substantial CO ₂
High	\$7,024,000,000	\$6,566,000,000	\$7,987,000,000
Restrained	\$4,816,000,000	\$5,345,000,000	\$6,800,000,000
Moderate	\$2,911,000,000	\$4,278,000,000	\$4,982,000,000
Low	\$1,265,000,000	\$3,541,000,000	\$4,061,000,000

Positive number means the Facility is less costly than the gas-fired CC alternative.

The Company continues to believe equal weighting of these scenario outcomes is most appropriate given the difficulty in assessing the outcome of a vast range of key variables such as future environmental regulations, possible climate change regulation, fuel prices, demand levels, potential federal portfolio requirements, federal policies toward new nuclear, the breadth and rate of expansion of new nuclear in the United States, and the interplay of other market forces. As such, the weighted average expected value of the relative savings for completion of the Facility as compared to the gas-fired CC alternative is \$5.0 billion based on the results provided in Table 14.1.

Alternatively, the results of the updated economic evaluation can be expressed in terms of the “breakeven capital cost to complete.” Table 14.2 below shows the results of the breakeven analysis that calculates the maximum capital expenditure that could be spent to complete the Facility and maintain lifetime costs that are equal to the cost of the gas-fired CC alternative. In all of the scenarios, the maximum capital cost to complete the Facility exceeds the Company’s current estimate of the cost to complete the Facility (including marginal construction financing costs) of \$2.7 billion.

Table 14.2

Relative Savings of the Facility versus CC as of February 28, 2014
“Break-Even Cost to Complete”

(In 2016 Dollars)

(Maximum Capital Costs to Complete the Facility and Remain Economic)

Fuel \ CO ₂	Existing CO ₂	Moderate CO ₂	Substantial CO ₂
High	\$7,746,000,000	\$7,419,000,000	\$8,435,000,000
Restrained	\$6,169,000,000	\$6,548,000,000	\$7,586,000,000
Moderate	\$4,809,000,000	\$5,785,000,000	\$6,288,000,000
Low	\$3,633,000,000	\$5,259,000,000	\$5,630,000,000

If the value is higher than the current estimated cost to complete of \$2.7 billion of in-service and construction financing costs, the Facility benefits customers. On an expected value basis, the Company’s results indicate that the cost to complete the Facility could increase by \$3.5 billion over the current estimated cost to complete the Facility before becoming uneconomic. (This value can be derived by averaging the results from the twelve scenarios above and then subtracting the current estimated cost to complete).

The analyses provided in Tables 14.1 and 14.2 are based on an economic assessment from an “incremental cost to complete” perspective, which ignores any potential cancellation fees or other costs that would be incurred if the project were stopped, as well as any fully-committed construction costs that would not be avoidable in the event the project is cancelled. If the results from the incremental cost to complete evaluation showed it was no longer cost-effective to pursue completing the Facility, a second cancellation assessment would be performed to determine the economic value of canceling the Facility. A cancellation assessment can provide the most appropriate perspective for deciding whether to cancel the Facility as it would include the impacts of any cancellation fees or other costs associated with cancelling the Facility in the economic analysis. However, because Tables 14.1 and 14.2 both reflect significant savings and benefits to customers from the incremental cost to complete perspective across a wide range of possible future fuel and carbon prices, a cancellation assessment is not warranted at this time.

In the Eighth VCM proceeding, the Commission ordered that delay scenarios of 24, 36 and 48 months be performed using the latest COD dates for the Units in future VCM filings. The Company has performed economic analysis in which the in-service dates are delayed by 24, 36 and 48 months from December 2017 and December 2018 for Units 3 and 4, respectively. These scenarios include additional capital costs and financing costs related to the delay scenarios, and the results are provided in Table 14.3, 14.4 and 14.5.

Table 14.3

**Relative Savings of the Facility versus CC as of February 28, 2014
December 2019 / December 2020 In-service (24 Month Delay) Scenario
“Break-Even Cost to Complete”**

(In 2016 Dollars)

(Maximum Capital Costs to Complete the Facility and Remain Economic)

Fuel \ CO ₂	Existing CO ₂	Moderate CO ₂	Substantial CO ₂
High	\$7,771,000,000	\$7,395,000,000	\$8,417,000,000
Restrained	\$6,235,000,000	\$6,541,000,000	\$7,590,000,000
Moderate	\$4,949,000,000	\$5,814,000,000	\$6,384,000,000
Low	\$3,796,000,000	\$5,260,000,000	\$5,712,000,000

If the value is higher than this scenario’s estimated cost to complete of \$3.4 billion of in-service and construction financing costs, the Facility benefits customers.

Table 14.4

**Relative Savings of the Facility versus CC as of February 28, 2014
December 2020 / December 2021 In-service (36 Month Delay) Scenario
“Break-Even Cost to Complete”**

(In 2016 Dollars)

(Maximum Capital Costs to Complete the Facility and Remain Economic)

Fuel \ CO ₂	Existing CO ₂	Moderate CO ₂	Substantial CO ₂
High	\$7,686,000,000	\$7,301,000,000	\$8,322,000,000
Restrained	\$6,182,000,000	\$6,459,000,000	\$7,512,000,000
Moderate	\$4,938,000,000	\$5,753,000,000	\$6,346,000,000
Low	\$3,798,000,000	\$5,188,000,000	\$5,665,000,000

If the value is higher than this scenario’s estimated cost to complete of \$3.75 billion of in-service and construction financing costs, the Facility benefits customers.

Table 14.5

Relative Savings of the Facility versus CC as of February 28, 2014
December 2021 / December 2022 In-service (48 Month Delay) Scenario
“Break-Even Cost to Complete”
(In 2016 Dollars)
(Maximum Capital Costs to Complete the Facility and Remain Economic)

Fuel \ CO ₂	Existing CO ₂	Moderate CO ₂	Substantial CO ₂
High	\$7,633,000,000	\$7,240,000,000	\$8,244,000,000
Restrained	\$6,159,000,000	\$6,414,000,000	\$7,445,000,000
Moderate	\$4,957,000,000	\$5,725,000,000	\$6,321,000,000
Low	\$3,834,000,000	\$5,150,000,000	\$5,630,000,000

If the value is higher than this scenario’s estimated cost to complete of \$4.1 billion of in-service and construction financing costs, the Facility benefits customers.

Economic Analysis Conclusion / Summary of Results

In summary, all scenario studies, with the exception of the Low Fuel/Existing CO₂ in the 48 Month Delay Scenario, indicate that the Facility would remain economic despite the additional costs associated with the delay scenarios. In the delay scenarios, the Facility remains less costly than the next best fuel alternative and will continue to benefit customers. These scenarios do not represent the Company’s projection for the ultimate outcome of the project but instead represent the delay scenarios ordered by the Commission in the Eighth VCM proceeding.

15. **The Company will be under a continuing obligation to supplement its response to PIA Staff DR STF-TN-1-2 by ensuring that the financing data reflected in the schedules attached to that DR response reflect the most current and updated information at the time of each semi-annual monitoring report. In addition, the Company will provide the most current information shared with each of the Rating Agencies.**

Simultaneous with this filing, the Company has filed supplemental PIA Staff DR STF-TN-1-2, and has included in that filing the most current information shared with each of the Rating Agencies.

The Company recommends removing this item in future VCM reports with the understanding that supplemental responses to STF-TN-1-2 will continue to be filed simultaneously with the VCM Reports.